Preparing data for ML models.

Encoding: Process of converting a categorical columns to numerical / vector representation.

Why? - Algorithms cannot process categorical data directly as they except fixed- numerical vector as input. Also, they use mathematical operation that are not defined for non-numerical data / string.

Types: One hot encoding & Label encoding.

One hot encoding: Use when categories are Nominal. This converts each unique category in a categorical column into a new binary column.

$$\# \text{ columns} = (n\text{-class} - 1)$$

GENDER
MALE
FEMALE
FEMALE
MALE
FEMALE
MALE

GENDER_MALE	~~GENDER_FEMALE~~
1	~~0~~
0	~~1~~
0	~~1~~
1	~~0~~
0	~~1~~
1	~~0~~

COUNTRY
INDIA
RUSSIA
CHINA
CHINA
INDIA
INDIA
RUSSIA

COUNTRY_INDIA	COUNTRY_RUSSIA	COUNTRY_CHINA
1	0	0
0	1	0
0	0	1
0	0	1
1	0	0
1	0	0
0	1	0

PYTHON:

1. pd.get_dummies
2. sklearn.preprocessing.OneHotEncoder

Label Encoding: Label encoding assigns a unique integer to each category in the dataset. It is used for both ordinal / nominal data.

The reason for using label encoding for nominal data is because its simple and memory efficient (this works only on algorithms that naturally handles ordinal relationships). The reason for using label encoding for ordinal data is to naturally assign higher weight to the classes of a column based on its importance.

Nominal

COUNTRY	COUNTRY
INDIA	1
RUSSIA	2
CHINA	3
CHINA	3
INDIA	1
INDIA	1
RUSSIA	2

Ordinal

POSITION	POSITION
Sn.Engineer	2
Jn.Engineer	1
Manager	4
TL	3
Jn.Engineer	1
Jn.Engineer	1
Sn.Engineer	2

PYTHON : sklearn.preprocessing.LabelEncoder, pd.map